



CIS422-Human Computer Interaction - Project Proposal

Project Group Number (in Blackboard): **Group 1**



Project Title: *FoodSaver*

Leader	ID	Member Name	Email Address
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System Description:

FoodSaver is a smart home-based mobile application for reducing food waste and efficiently managing the food stock of the household. It is based on a simple, interactive, and user-friendly interface for the user to track their food stock and monitor the expiration dates using a visual calendar similar to the calendar used by smartphones. This application also ensures timely notifications for the user about the expiring products so that they do not forget to consume the products and reduce the chances of unnecessary wastage.

In order to simplify the interaction with the application, FoodSaver does not require the user to type the ingredients. It simply uses checkboxes for selecting the ingredients from the predefined categories such as vegetables, fruits, meat, poultry products, dairy products, grains, and other pantry items using visual icons.

Based on the selected food materials, the system will smartly suggest relevant recipes and fast meal ideas. This will assist users in determining what to cook next and at the same time make the most out of the available food materials. Moreover, FoodSaver includes a community component that enables users to share recipes and share their views and experience in cooking. The recipes are reviewed by experienced home cooks.

The FoodSaver application is built using fundamental principles of Human-Computer Interaction (HCI) that include usability, simplicity, efficiency, cognitive load reduction, and user-centeredness.

System Functionality:

Our app tackles the problem of food waste at home with four main features:

1. Keep Track of Your Food

Users can add the food they have at home along with its expiry date into a simple, calendar-like view. This makes it easy to see what you have and when it will go bad.

2. Get Reminders Before Food Expires

The app automatically sends alerts to your phone before your food items expire. This helps you remember to use them in time, so nothing gets thrown away unnecessarily.

3. Get Recipe Ideas Based on What You Have

Users can check off ingredients they already own from organized categories (like veggies, dairy, meat). The app then suggests recipes they can make right away, helping them use up food before it spoils.

4. Share and Discover Recipes with Others

Users can share their own recipes, rate others, and see what people are cooking. To make sure recipes are good, they are checked and approved by experienced home cooks before everyone can see them.

Constraints

1. Limited Internet Connectivity
 - Certain aspects of this application, like sharing recipes and looking at community recipes, require an internet connection, which can be limited without a network connection.
2. Accuracy of User Input
 - The system relies on users to input expiration dates correctly. Additionally, they have to choose ingredients. This could influence the accuracy of notifications and recipe advice.
3. Limited Recipe Database
 - Recipe suggestions are based on the availability of recipes within the application, and this might be limited, especially in the early stages.
4. Variation in Users' Dietary Preferences
 - The suggested recipes might not be appropriate for all users, as people are of different cultures, dietary habits, allergies, or their dietary restrictions.
5. Mobile Notification Dependency
 - The reminder delivery time is dependent upon the settings of the phone's notification feature. In the event that the feature is either disabled or restricted, the users might not receive key notifications.
6. Prototype Limitation
 - Most likely, the application may not have the latest features, such as integrating the stores itself or tracking food through artificial intelligence.